OPT · EXERCISE SHEET 3 · SOLUTIONS

The simplex method

Step-by-step solutions to the 6 exercises of Série d'exercices N°3. USTHB, M1 Master Big Data, Optimisation Linéaire, 2026/2027.

How every solution works

0 · Set up

Add one leftover (slack) variable per ≤ limit to get equations. Start at the corner “everything 0”: the leftovers form the basis.

Step 1 · What to increase

In the Z row, take the biggest positive number. That variable enters (it stops being 0).

Tie: do step 2 for each, pick the bigger (Z number × smallest ratio).

Step 2 · How far

For each row with a positive number in that column: cste ÷ number. The smallest wins; that row's variable leaves (drops to 0). Rows with 0 or a negative number are skipped.

Step 3 · Rewrite

Divide the pivot row by the pivot. In every other row (Z too), subtract (its number in the column) × new pivot row.

Read the corner

Each variable in the basis column = its cste. All others = 0. Profit = Z-row cste without its minus.

Stop

When the Z row has no positive number. A 0 under a variable outside the basis means another corner has the same profit.

Notation: the table's last column “cste” holds the right-hand sides; in the Z row it holds −Z (minus the current profit). L_p = new pivot row. Every table, ratio and answer was computed by program with exact fractions and checked against every corner of the feasible region.

Exercise 1 Two products, three limits

ASKED

Solve (P): max $Z = 10x_1 + 15x_2$ subject to $5x_1 + 2x_2 \le 80$, $x_1 + x_2 \le 20$, $x_1 + 2x_2 \le 30$, $x_1, x_2 \ge 0$, with the simplex.

METHOD

Slacks, start at (0, 0), repeat steps 1–3 until the Z row has no positive number.

1 Standard form

Each limit gets a leftover: x_3, x_4, x_5.

$$5x_1 + 2x_2 + x_3 = 80$$
$$x_1 + x_2 + x_4 = 20$$
$$x_1 + 2x_2 + x_5 = 30$$
$$Z = 10x_1 + 15x_2 \rightarrow \text{maximise} \cdot \text{all variables} \ge 0$$

2 Starting table

Starting table

basis	x_1	x_2	x_3	x_4	x_5	cste
x_3	5	2	1	0	0	80
x_4	1	1	0	1	0	20
x_5	1	2	0	0	1	30
Z	10	15	0	0	0	0

Read the corner. In the basis column: $x_3 = 80$, $x_4 = 20$, $x_5 = 30$. Not in the basis, so equal to 0: x_1, x_2.

Corner (x_1, x_2) = **(0, 0)**, profit Z = **0** (the Z-row cste 0 without its minus sign).

3 Round 1: x_2 enters, x_5 leaves

ROUND 1

Step 1 **What to increase?**

Z row (for variables not in the basis): 10 under x_1, 15 under x_2. The biggest positive number is **15**, under x_2: one more unit of x_2 adds 15 to the profit, the most. So **x_2 enters** (it will stop being 0).

Step 2 **How far can x_2 go?**

For each row with a positive number in the x_2 column: cste ÷ that number = how much x_2 fits before that row's variable reaches 0.

row	cste	x_2 column	ratio
x_3	80	2	80 ÷ 2 = **40**
x_4	20	1	20 ÷ 1 = **20**
x_5	30	2	30 ÷ 2 = **15** ← smallest

The smallest ratio is **15**, in the x_5 row: **x_5 leaves** (it drops to 0). Pivot = the number where the x_2 column meets the x_5 row = **2**.

Table before the rewrite: x_2 column shaded, x_5 row outlined, pivot boxed

basis	x_1	x_2	x_3	x_4	x_5	cste
x_3	5	2	1	0	0	80
x_4	1	1	0	1	0	20
x_5	1	2	0	0	1	30
Z	10	15	0	0	0	0

Step 3 **Rewrite the table**

(a) Divide the x_5 row by the pivot 2; it becomes the x_2 row. (b) In every other row, subtract (its number in the x_2 column) × the new pivot row, so the x_2 column becomes 1 in its own row and 0 everywhere else.

Table after round 1

basis	x_1	x_2	x_3	x_4	x_5	cste	how this row was computed
x_3	4	0	1	0	-1	50	old x_3 row − 2·new pivot row
x_4	1/2	0	0	1	-1/2	5	old x_4 row − new pivot row
x_2	1/2	1	0	0	1/2	15	old x_5 row ÷ 2
Z	5/2	0	0	0	-15/2	-225	old Z row − 15·new pivot row

Read the corner. In the basis column: x_3 = 50, x_4 = 5, x_2 = 15. Not in the basis, so equal to 0: x_1, x_5.

Corner (x_1, x_2) = **(0, 15)**, profit Z = **225** (the Z-row cste −225 without its minus sign).

4 Round 2: x_1 enters, x_4 leaves

ROUND 2

Step 1 **What to increase?**

Z row (for variables not in the basis): 5/2 under x_1, −15/2 under x_5. The biggest positive number is **5/2**, under x_1: one more unit of x_1 adds 5/2 to the profit, the most. So **x_1 enters** (it will stop being 0).

Step 2 **How far can x_1 go?**

For each row with a positive number in the x_1 column: cste ÷ that number = how much x_1 fits before that row's variable reaches 0.

row	cste	x_1 column	ratio
x_3	50	4	50 ÷ 4 = **25/2**
x_4	5	1/2	5 ÷ (1/2) = **10** ← smallest
x_2	15	1/2	15 ÷ (1/2) = **30**

The smallest ratio is **10**, in the x_4 row: **x_4 leaves** (it drops to 0). Pivot = the number where the x_1 column meets the x_4 row = **1/2**.

Table before the rewrite: x_1 column shaded, x_4 row outlined, pivot boxed

basis	x_1	x_2	x_3	x_4	x_5	cste
x_3	4	0	1	0	-1	50
x_4	1/2	0	0	1	-1/2	5
x_2	1/2	1	0	0	1/2	15
Z	5/2	0	0	0	-15/2	-225

Step 3 **Rewrite the table**

(a) Divide the x_4 row by the pivot 1/2; it becomes the x_1 row. (b) In every other row, subtract (its number in the x_1 column) × the new pivot row, so the x_1 column becomes 1 in its own row and 0 everywhere else.

Table after round 2

basis	x_1	x_2	x_3	x_4	x_5	cste	how this row was computed
x_3	0	0	1	-8	3	10	old x_3 row − 4·new pivot row
x_1	1	0	0	2	-1	10	old x_4 row ÷ (1/2)
x_2	0	1	0	-1	1	10	old x_2 row − (1/2)·new pivot row
Z	0	0	0	-5	-5	-250	**old Z row − (5/2)·new pivot row**

Read the corner. In the basis column: x_3 = 10, x_1 = 10, x_2 = 10. Not in the basis, so equal to 0: x_4, x_5.

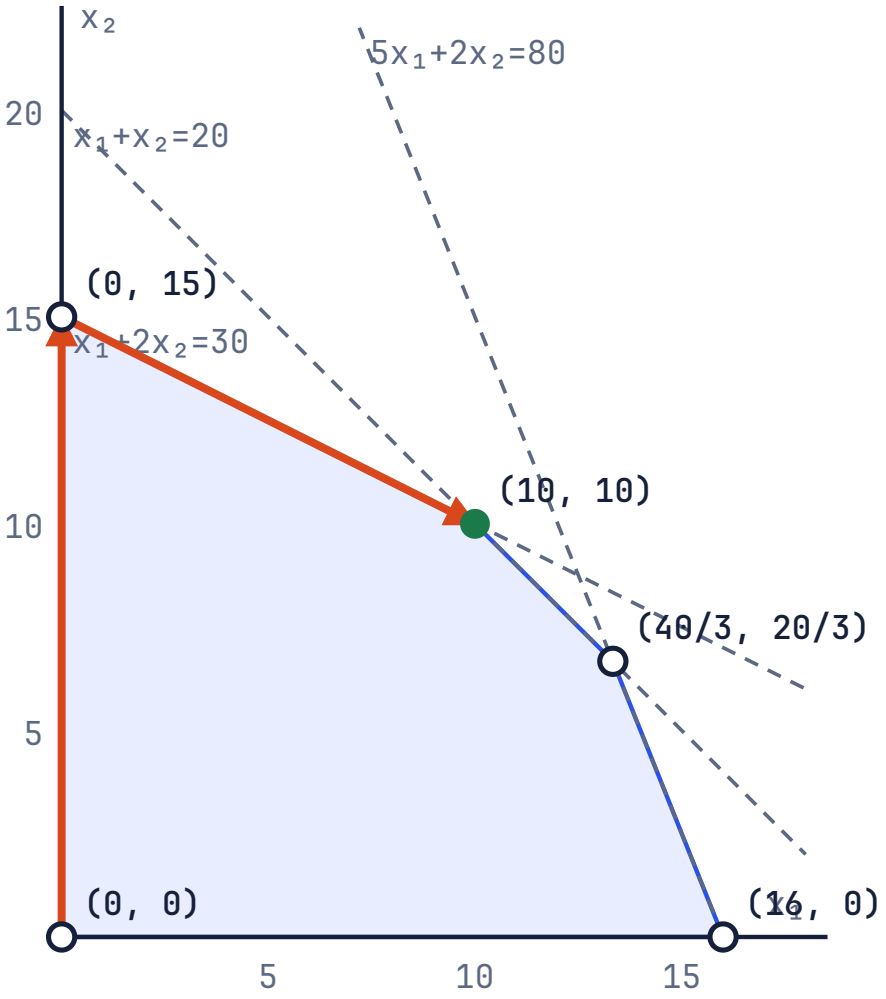
Corner (x_1, x_2) = **(10, 10)**, profit Z = **250** (the Z-row cste −250 without its minus sign).

5 Stop? Read the answer

Z row (variables not in the basis): −5 under x_4, −5 under x_5. **No positive number**: any change would lower the profit, so we stop. This is the best plan.

No 0 under a variable outside the basis, so this best plan is the only one.

6 Check on the graph



Red: the simplex path (0, 0) → (0, 15) → (10, 10). Green: the best corner.

corner	(10, 10)	(40/3, 20/3)	(0, 15)	(16, 0)	(0, 0)
$Z = 10x_1 + 15x_2$	250	700/3	225	160	0

Every corner was tried: (10, 10) gives the most.

ANSWER

$x_1 = 10$, $x_2 = 10$, Z max = 250, after 2 rounds. Leftovers: $x_3 = 10$ (first limit not full), $x_4 = x_5 = 0$ (second and third limits are full).

Exercise 2 Three products

ASKED

Solve (P): max $Z = 120x_1 + 108x_2 + 75x_3$ subject to $x_1 + x_2 + x_3 \le 12$, $x_1 \le 5$, $8x_1 + 7x_2 + 5x_3 \le 145$, all ≥ 0.

METHOD

Same three steps. With three variables there is no graph; the table does all the work.

1 Standard form

Leftovers x_4, x_5, x_6, one per limit.

$$x_1 + x_2 + x_3 + x_4 = 12$$
$$x_1 + x_5 = 5$$
$$8x_1 + 7x_2 + 5x_3 + x_6 = 145$$
$$Z = 120x_1 + 108x_2 + 75x_3 \rightarrow \text{maximise} \cdot \text{all variables} \ge 0$$

2 Starting table

Starting table

basis	x_1	x_2	x_3	x_4	x_5	x_6	cste
x_4	1	1	1	1	0	0	12
x_5	1	0	0	0	1	0	5
x_6	8	7	5	0	0	1	145
Z	120	108	75	0	0	0	0

Read the corner. In the basis column: $x_4 = 12$, $x_5 = 5$, $x_6 = 145$. Not in the basis, so equal to 0: x_1, x_2, x_3.

Corner (x_1, x_2, x_3) = **(0, 0, 0)**, profit Z = **0** (the Z-row cste 0 without its minus sign).

3 Round 1: x_1 enters, x_5 leaves

ROUND 1

Step 1 What to increase?

Z row (for variables not in the basis): 120 under x_1, 108 under x_2, 75 under x_3. The biggest positive number is **120**, under x_1: one more unit of x_1 adds 120 to the profit, the most. So **x_1 enters** (it will stop being 0).

Step 2 How far can x_1 go?

For each row with a positive number in the x_1 column: cste ÷ that number = how much x_1 fits before that row's variable reaches 0.

row	cste	x_1 column	ratio
x_4	12	1	12 ÷ 1 = **12**
x_5	5	1	5 ÷ 1 = **5** ← smallest
x_6	145	8	145 ÷ 8 = **145/8**

The smallest ratio is **5**, in the x_5 row: **x_5 leaves** (it drops to 0). Pivot = the number where the x_1 column meets the x_5 row = **1**.

Table before the rewrite: x_1 column shaded, x_5 row outlined, pivot boxed

basis	x_1	x_2	x_3	x_4	x_5	x_6	cste
x_4	1	1	1	1	0	0	12
x_5	1	0	0	0	1	0	5
x_6	8	7	5	0	0	1	145
Z	120	108	75	0	0	0	0

Step 3 **Rewrite the table**

(a) Divide the x_5 row by the pivot 1; it becomes the x_1 row. (b) In every other row, subtract (its number in the x_1 column) × the new pivot row, so the x_1 column becomes 1 in its own row and 0 everywhere else.

Table after round 1

basis	x_1	x_2	x_3	x_4	x_5	x_6	cste	how this row was computed
x_4	0	1	1	1	-1	0	7	old x_4 row − new pivot row
x_1	1	0	0	0	1	0	5	old x_5 row ÷ 1
x_6	0	7	5	0	-8	1	105	old x_6 row − 8·new pivot row
Z	0	108	75	0	-120	0	-600	old Z row − 120·new pivot row

Read the corner. In the basis column: x_4 = 7, x_1 = 5, x_6 = 105. Not in the basis, so equal to 0: x_2, x_3, x_5.

Corner (x_1, x_2, x_3) = **(5, 0, 0)**, profit Z = **600** (the Z-row cste −600 without its minus sign).

4 Round 2: x_2 enters, x_4 leaves

ROUND 2

Step 1 **What to increase?**

Z row (for variables not in the basis): 108 under x_2, 75 under x_3, −120 under x_5. The biggest positive number is **108**, under x_2: one more unit of x_2 adds 108 to the profit, the most. So **x_2 enters** (it will stop being 0).

Step 2 **How far can x_2 go?**

For each row with a positive number in the x_2 column: cste ÷ that number = how much x_2 fits before that row's variable reaches 0.

row	cste	x_2 column	ratio
x_4	7	1	7 ÷ 1 = **7** ← smallest
x_1	5	0	skip: 0 in the column (never runs out)
x_6	105	7	105 ÷ 7 = **15**

The smallest ratio is **7**, in the x_4 row: **x_4 leaves** (it drops to 0). Pivot = the number where the x_2 column meets the x_4 row = **1**.

Table before the rewrite: x_2 column shaded, x_4 row outlined, pivot boxed

basis	x_1	x_2	x_3	x_4	x_5	x_6	cste
x_4	0	1	1	1	-1	0	7
x_1	1	0	0	0	1	0	5
x_6	0	7	5	0	-8	1	105
Z	0	108	75	0	-120	0	-600

Step 3 **Rewrite the table**

(a) Divide the x_4 row by the pivot 1; it becomes the x_2 row. (b) In every other row, subtract (its number in the x_2 column) × the new pivot row, so the x_2 column becomes 1 in its own row and 0 everywhere else.

Table after round 2

basis	x_1	x_2	x_3	x_4	x_5	x_6	cste	how this row was computed
x_2	0	1	1	1	-1	0	7	old x_4 row ÷ 1
x_1	1	0	0	0	1	0	5	copied (0 in the column)
x_6	0	0	-2	-7	-1	1	56	old x_6 row − 7·new pivot row
Z	0	0	-33	-108	-12	0	-1356	old Z row − 108·new pivot row

Read the corner. In the basis column: x_2 = 7, x_1 = 5, x_6 = 56. Not in the basis, so equal to 0: x_3, x_4, x_5.

Corner (x_1, x_2, x_3) = **(5, 7, 0)**, profit Z = **1356** (the Z-row cste −1356 without its minus sign).

5 Stop? Read the answer

Z row (variables not in the basis): −33 under x_3, −108 under x_4, −12 under x_5. **No positive number**: any change would lower the profit, so we stop. This is the best plan.

No 0 under a variable outside the basis, so this best plan is the only one.

6 Check

Plug (5, 7, 0) into the limits: 5 + 7 + 0 = 12 ≤ 12 ✓ (full), 5 ≤ 5 ✓ (full), 8·5 + 7·7 + 5·0 = 89 ≤ 145 ✓ (56 left = x_6). Profit 120·5 + 108·7 = 600 + 756 = **1356**. Trying every corner of the region gives the same maximum.

ANSWER

x_1 = 5, x_2 = 7, x_3 = 0, Z max = 1356, after 2 rounds. The first two limits are full; 56 units of the third are left (x_6 = 56).

Exercise 3 — A ≥ limit and several best plans

ASKED

Solve (P): max $Z = x_1 - x_2$ subject to $2x_1 - x_2 \geq -4$, $x_1 - x_2 \leq 4$, $x_1 + x_2 \leq 10$, $x_1, x_2 \geq 0$.

METHOD

Turn the ≥ limit into a ≤ by multiplying by −1, then the usual three steps. Watch the 0 in the last Z row.

1 Standard form

The first limit is $2x_1 - x_2 \geq -4$. Multiply both sides by −1 (this flips ≥ into ≤): **$-2x_1 + x_2 \leq 4$**. Now every limit is ≤ with a positive right-hand side, so (0, 0) is allowed and the leftovers can start the table.

$$-2x_1 + x_2 + x_3 = 4$$
$$x_1 - x_2 + x_4 = 4$$
$$x_1 + x_2 + x_5 = 10$$
$$Z = x_1 - x_2 \rightarrow \text{maximise} \cdot \text{all variables} \geq 0$$

2 Starting table

Starting table

basis	x_1	x_2	x_3	x_4	x_5	cste
x_3	-2	1	1	0	0	4
x_4	1	-1	0	1	0	4
x_5	1	1	0	0	1	10
Z	1	-1	0	0	0	0

Read the corner. In the basis column: $x_3 = 4$, $x_4 = 4$, $x_5 = 10$. Not in the basis, so equal to 0: x_1, x_2.

Corner (x_1, x_2) = **(0, 0)**, profit Z = **0** (the Z-row cste 0 without its minus sign).

3 Round 1: x_1 enters, x_4 leaves

ROUND 1

Step 1 — What to increase?

Z row (for variables not in the basis): 1 under x_1, −1 under x_2. The biggest positive number is **1**, under x_1: one more unit of x_1 adds 1 to the profit, the most. So **x_1 enters** (it will stop being 0).

Step 2 — How far can x_1 go?

For each row with a positive number in the x_1 column: cste ÷ that number = how much x_1 fits before that row's variable reaches 0.

row	cste	x_1 column	ratio
x_3	4	-2	skip: negative number (-2), so x_3 grows
x_4	4	1	4 ÷ 1 = **4** ← smallest
x_5	10	1	10 ÷ 1 = **10**

The smallest ratio is **4**, in the x_4 row: **x_4 leaves** (it drops to 0). Pivot = the number where the x_1 column meets the x_4 row = **1**.

Table before the rewrite: x_1 column shaded, x_4 row outlined, pivot boxed

basis	x_1	x_2	x_3	x_4	x_5	cste
x_3	-2	1	1	0	0	4
x_4	1	-1	0	1	0	4
x_5	1	1	0	0	1	10
Z	1	-1	0	0	0	0

Step 3 **Rewrite the table**

(a) Divide the x_4 row by the pivot 1; it becomes the x_1 row. (b) In every other row, subtract (its number in the x_1 column) × the new pivot row, so the x_1 column becomes 1 in its own row and 0 everywhere else.

Table after round 1

basis	x_1	x_2	x_3	x_4	x_5	cste	how this row was computed
x_3	0	-1	1	2	0	12	old x_3 row + 2·new pivot row
x_1	1	-1	0	1	0	4	old x_4 row ÷ 1
x_5	0	2	0	-1	1	6	old x_5 row − new pivot row
Z	0	0	0	-1	0	-4	**old Z row − new pivot row**

Read the corner. In the basis column: x_3 = 12, x_1 = 4, x_5 = 6. Not in the basis, so equal to 0: x_2, x_4.

Corner (x_1, x_2) = **(4, 0)**, profit Z = **4** (the Z-row cste −4 without its minus sign).

4 Stop? Read the answer

Z row (variables not in the basis): 0 under x_2, −1 under x_4. **No positive number**: any change would lower the profit, so we stop. This is the best plan.

There is a 0 under x_2: another corner gives the same profit (see below).

5 Extra round: x_2 enters, x_5 leaves (same profit)

ROUND 2

Step 1 **What to increase?**

The Z row (0 under x_2, −1 under x_4) has **no positive number**, so this corner is already a best plan. But x_2 is not in the basis and has **0** in the Z row: increasing it changes nothing in the profit. So we do one more round with x_2 to find **another corner with the same profit**.

Step 2 **How far can x_2 go?**

For each row with a positive number in the x_2 column: cste ÷ that number = how much x_2 fits before that row's variable reaches 0.

row	cste	x_2 column	ratio
x_3	12	-1	skip: negative number (-1), so x_3 grows
x_1	4	-1	skip: negative number (-1), so x_1 grows
x_5	6	2	6 ÷ 2 = **3** ← smallest

The smallest ratio is **3**, in the x_5 row: **x_5 leaves** (it drops to 0). Pivot = the number where the x_2 column meets the x_5 row = **2**.

Table before the rewrite: x_2 column shaded, x_5 row outlined, pivot boxed

basis	x_1	x_2	x_3	x_4	x_5	cste
x_3	0	-1	1	2	0	12
x_1	1	-1	0	1	0	4
x_5	0	2	0	-1	1	6
Z	0	0	0	-1	0	-4

Step 3 Rewrite the table

(a) Divide the x_5 row by the pivot 2; it becomes the x_2 row. (b) In every other row, subtract (its number in the x_2 column) × the new pivot row, so the x_2 column becomes 1 in its own row and 0 everywhere else.

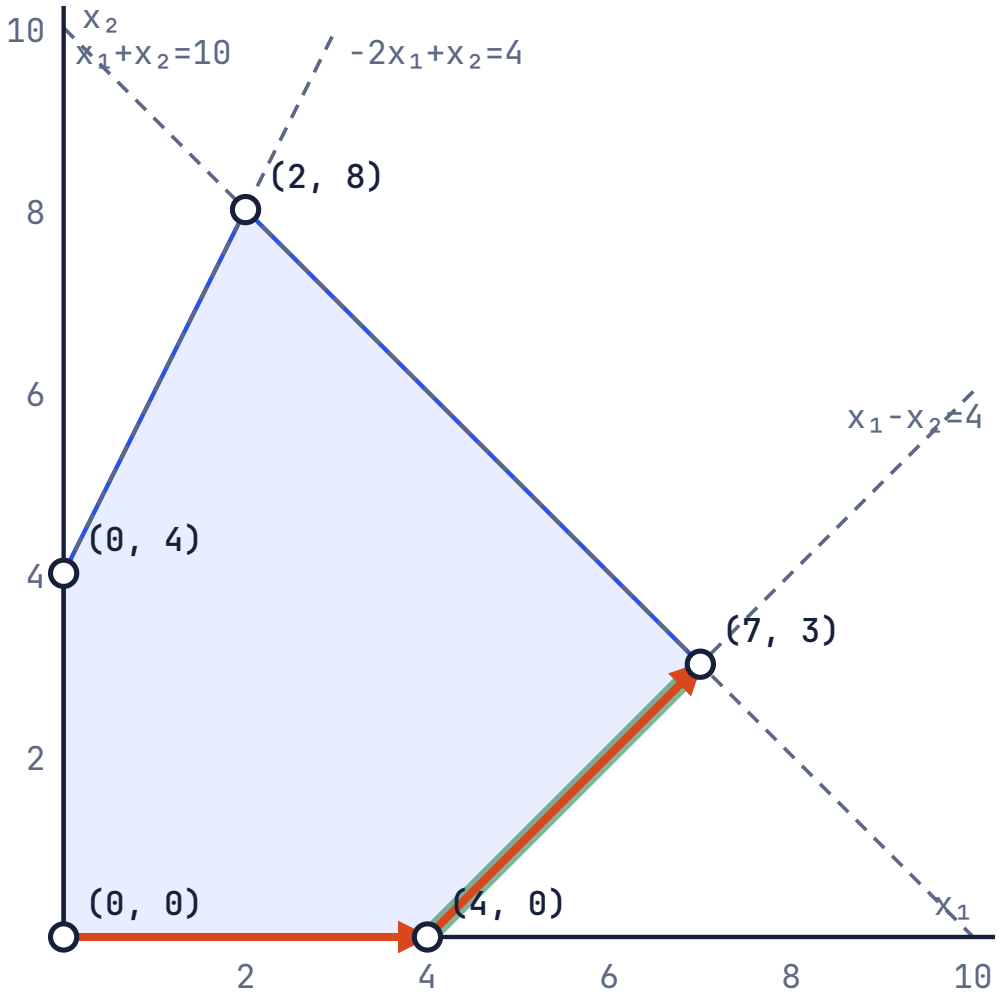
Table after round 2

basis	x_1	x_2	x_3	x_4	x_5	cste	how this row was computed
x_3	0	0	1	3/2	1/2	15	old x_3 row + new pivot row
x_1	1	0	0	1/2	1/2	7	old x_1 row + new pivot row
x_2	0	1	0	-1/2	1/2	3	old x_5 row ÷ 2
Z	0	0	0	-1	0	-4	**copied (0 in the column)**

Read the corner. In the basis column: x_3 = 15, x_1 = 7, x_2 = 3. Not in the basis, so equal to 0: x_4, x_5.

Corner (x_1, x_2) = **(7, 3)**, profit Z = **4** (the Z-row cste −4 without its minus sign).

6 Check on the graph



The side from (4, 0) to (7, 3) lies on $x_1 - x_2 = 4$, so every point on it has $Z = x_1 - x_2 = 4$ (green).

corner	(4, 0)	(7, 3)	(0, 0)	(0, 4)	(2, 8)
$Z = x_1 - x_2$	4	4	0	-4	-6

Two corners tie at Z = 4: (4, 0) and (7, 3).

ANSWER

Z max = 4. It is reached at (4, 0) and at (7, 3), and at every point of the side between them: $x_1 - x_2 = 4$ with $4 \le x_1 \le 7$ (for example (5, 1) or (6, 2)).

Exercise 4 Starting from a given corner

ASKED

(P): max $Z = 10x_1 + 14x_2$ subject to $x_1 + x_2 \le 12$, $x_1 \ge 8$, $x_2 \le 6$, $x_1, x_2 \ge 0$. 1) Show (12, 0) is a corner, write the standard form and the basic solution x_B at (12, 0). 2) Solve with the simplex starting from (12, 0).

METHOD

Check the two limits that meet at (12, 0). Use a surplus variable for the ≥ limit. Rewrite the table so x_1, x_4, x_5 form the basis, then the usual steps.

1 Question 1: (12, 0) is a corner; standard form; basic solution

Is (12, 0) a corner? A corner is where two limits are exactly reached and every limit is respected.

- $x_1 + x_2 = 12 + 0 = 12$: reached exactly.
- $x_2 = 0$: the sign limit $x_2 \ge 0$ is reached exactly.
- The other limits hold: $x_1 = 12 \ge 8$ ✓, $x_2 = 0 \le 6$ ✓, $x_1 \ge 0$ ✓.

The two lines $x_1 + x_2 = 12$ and $x_2 = 0$ cross at exactly one point, (12, 0), and it is allowed. So it is a corner (sommet).

Standard form. ≤ limits get a leftover (+); the ≥ limit gets a **surplus** (−), the amount above the minimum:

```
x1 + x2 + x3 = 12
x1 - x4 = 8
x2 + x5 = 6
Z = 10x1 + 14x2 → maximise · all variables ≥ 0
```

x_3 = what is left of 12; x_4 = how much x_1 is above 8; x_5 = what is left of 6. Here we cannot start at (0, 0): $x_1 = 0$ would give $x_4 = -8 < 0$, not allowed. That is why the exercise gives (12, 0) as the start.

Basic solution at (12, 0). $x_1 = 12$, $x_2 = 0$ → $x_3 = 12 - 12 - 0 = 0$, $x_4 = 12 - 8 = 4$, $x_5 = 6 - 0 = 6$. Zero variables (out of the basis): x_2, x_3. In the basis: **$x_B = (x_1, x_4, x_5) = (12, 4, 6)$**.

2 Question 2: the starting table at (12, 0)

Write the table for this basis. Express x_1, x_4, x_5 and Z with the zero variables x_2, x_3 only:

```
x1 = 12 - x2 - x3 → row: x1 + x2 + x3 = 12
x4 = x1 - 8 = (12 - x2 - x3) - 8 = 4 - x2 - x3 → row: x2 + x3 + x4 = 4
x5 = 6 - x2 → row: x2 + x5 = 6
Z = 10x1 + 14x2 = 10(12 - x2 - x3) + 14x2 = 120 + 4x2 - 10x3 → Z row: 0, 4, -10, 0, 0 | -120
```

Starting table at (12, 0), profit 120

basis	x_1	x_2	x_3	x_4	x_5	cste
x_1	1	1	1	0	0	12
x_4	0	1	1	1	0	4
x_5	0	1	0	0	1	6
Z	0	4	-10	0	0	-120

3 Round 1: x_2 enters, x_4 leaves

ROUND 1

Step 1 What to increase?

Z row (for variables not in the basis): 4 under x_2, −10 under x_3. The biggest positive number is **4**, under x_2: one more unit of x_2 adds 4 to the profit, the most. So **x_2 enters** (it will stop being 0).

Step 2 How far can x_2 go?

For each row with a positive number in the x_2 column: cste ÷ that number = how much x_2 fits before that row's variable reaches 0.

row	cste	x_2 column	ratio
x_1	12	1	12 ÷ 1 = **12**
x_4	4	1	4 ÷ 1 = **4** ← smallest
x_5	6	1	6 ÷ 1 = **6**

The smallest ratio is **4**, in the x_4 row: **x_4 leaves** (it drops to 0). Pivot = the number where the x_2 column meets the x_4 row = **1**.

Table before the rewrite: x_2 column shaded, x_4 row outlined, pivot boxed

basis	x_1	x_2	x_3	x_4	x_5	cste
x_1	1	1	1	0	0	12
x_4	0	1	1	1	0	4
x_5	0	1	0	0	1	6
Z	0	4	-10	0	0	-120

Step 3 Rewrite the table

(a) Divide the x_4 row by the pivot 1; it becomes the x_2 row. (b) In every other row, subtract (its number in the x_2 column) × the new pivot row, so the x_2 column becomes 1 in its own row and 0 everywhere else.

Table after round 1

basis	x_1	x_2	x_3	x_4	x_5	cste	how this row was computed
x_1	1	0	0	-1	0	8	old x_1 row − new pivot row
x_2	0	1	1	1	0	4	old x_4 row ÷ 1
x_5	0	0	-1	-1	1	2	old x_5 row − new pivot row
Z	0	0	-14	-4	0	-136	old Z row − 4·new pivot row

Read the corner. In the basis column: x_1 = 8, x_2 = 4, x_5 = 2. Not in the basis, so equal to 0: x_3, x_4.

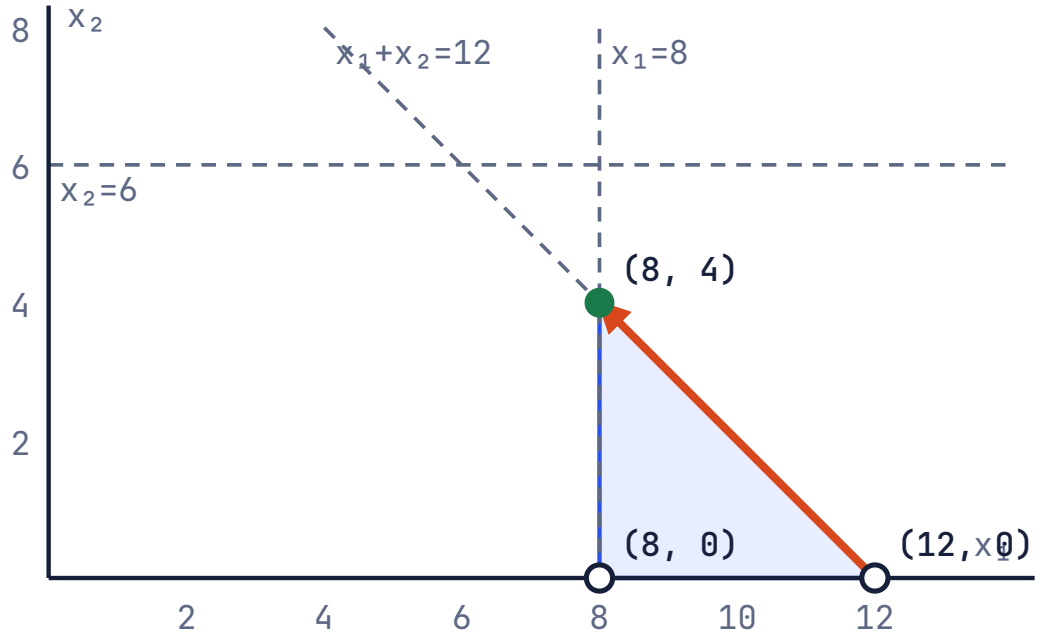
Corner (x_1, x_2) = **(8, 4)**, profit Z = **136** (the Z-row cste −136 without its minus sign).

4 Stop? Read the answer

Z row (variables not in the basis): −14 under x_3, −4 under x_4. **No positive number**: any change would lower the profit, so we stop. This is the best plan.

No 0 under a variable outside the basis, so this best plan is the only one.

5 Check on the graph



The allowed region is the small triangle. Red: from the start (12, 0) to the best corner (8, 4).

corner	(8, 4)	(12, 0)	(8, 0)
$Z = 10x_1 + 14x_2$	136	120	80

ANSWER

1) (12, 0) is a corner; $x_B = (x_1, x_4, x_5) = (12, 4, 6)$, with $x_2 = x_3 = 0$, $Z = 120$.

2) **$x_1 = 8$, $x_2 = 4$, Z max = 136**, after 1 round. $x_3 = 0$ and $x_4 = 0$ ($x_1 + x_2 = 12$ and $x_1 = 8$ are both reached), $x_5 = 2$.

Exercise 5 A tie in the first round

ASKED

Solve (P): max $Z = 5x_1 + 5x_2 + 3x_3$ subject to $x_1 + 3x_2 + x_3 \le 3$, $-x_1 + 3x_3 \le 2$, $2x_1 - x_2 + 2x_3 \le 4$, $2x_1 + 3x_2 - x_3 \le 2$, all ≥ 0.

METHOD

The usual three steps. In round 1, x_1 and x_2 both show 5: break the tie by the bigger gain. Fractions appear; keep them exact.

1 Standard form

Four limits, four leftovers x_4 to x_7.

$$x_1 + 3x_2 + x_3 + x_4 = 3$$
$$-x_1 + 3x_3 + x_5 = 2$$
$$2x_1 - x_2 + 2x_3 + x_6 = 4$$
$$2x_1 + 3x_2 - x_3 + x_7 = 2$$
$$Z = 5x_1 + 5x_2 + 3x_3 \rightarrow \text{maximise} \cdot \text{all variables} \ge 0$$

2 Starting table

Starting table

basis	x_1	x_2	x_3	x_4	x_5	x_6	x_7	cste
x_4	1	3	1	1	0	0	0	3
x_5	-1	0	3	0	1	0	0	2
x_6	2	-1	2	0	0	1	0	4
x_7	2	3	-1	0	0	0	1	2
Z	5	5	3	0	0	0	0	0

Read the corner. In the basis column: $x_4 = 3$, $x_5 = 2$, $x_6 = 4$, $x_7 = 2$. Not in the basis, so equal to 0: x_1, x_2, x_3.

Corner (x_1, x_2, x_3) = **(0, 0, 0)**, profit Z = **0** (the Z-row cste 0 without its minus sign).

3 Round 1: x_1 enters, x_7 leaves

ROUND 1

Step 1 **What to increase?**

Z row (for variables not in the basis): 5 under x_1, 5 under x_2, 3 under x_3. **Tie:** the biggest positive number, 5, appears more than once. Do step 2 for each tied variable and compare the profit each would bring: x_1: smallest ratio 1, so profit gained $5 \times 1 =$ **5**; x_2: smallest ratio 2/3, so profit gained $5 \times 2/3 =$ **10/3**. The bigger gain wins: **x_1 enters**.

Step 2 **How far can x_1 go?**

For each row with a positive number in the x_1 column: cste ÷ that number = how much x_1 fits before that row's variable reaches 0.

row	cste	x_1 column	ratio
x_4	3	1	3 ÷ 1 = **3**
x_5	2	-1	skip: negative number (-1), so x_5 grows
x_6	4	2	4 ÷ 2 = **2**
x_7	2	2	2 ÷ 2 = **1** ← smallest

The smallest ratio is **1**, in the x_7 row: **x_7 leaves** (it drops to 0). Pivot = the number where the x_1 column meets the x_7 row = **2**.

Table before the rewrite: x_1 column shaded, x_7 row outlined, pivot boxed

basis	x_1	x_2	x_3	x_4	x_5	x_6	x_7	cste
x_4	1	3	1	1	0	0	0	3
x_5	-1	0	3	0	1	0	0	2
x_6	2	-1	2	0	0	1	0	4
x_7	2	3	-1	0	0	0	1	2
Z	5	5	3	0	0	0	0	0

Step 3 **Rewrite the table**

(a) Divide the x_7 row by the pivot 2; it becomes the x_1 row. (b) In every other row, subtract (its number in the x_1 column) × the new pivot row, so the x_1 column becomes 1 in its own row and 0 everywhere else.

Table after round 1

basis	x_1	x_2	x_3	x_4	x_5	x_6	x_7	cste	how this row was computed
x_4	0	3/2	3/2	1	0	0	-1/2	2	old x_4 row − new pivot row
x_5	0	3/2	5/2	0	1	0	1/2	3	old x_5 row + new pivot row
x_6	0	-4	3	0	0	1	-1	2	old x_6 row − 2·new pivot row
x_1	1	3/2	-1/2	0	0	0	1/2	1	old x_7 row ÷ 2
Z	0	-5/2	11/2	0	0	0	-5/2	-5	**old Z row − 5·new pivot row**

Read the corner. In the basis column: x_4 = 2, x_5 = 3, x_6 = 2, x_1 = 1. Not in the basis, so equal to 0: x_2, x_3, x_7.

Corner (x_1, x_2, x_3) = **(1, 0, 0)**, profit Z = **5** (the Z-row cste −5 without its minus sign).

4 Round 2: x_3 enters, x_6 leaves

ROUND 2

Step 1 **What to increase?**

Z row (for variables not in the basis): −5/2 under x_2, 11/2 under x_3, −5/2 under x_7. The biggest positive number is **11/2**, under x_3: one more unit of x_3 adds 11/2 to the profit, the most. So **x_3 enters** (it will stop being 0).

Step 2 How far can x_3 go?

For each row with a positive number in the x_3 column: cste ÷ that number = how much x_3 fits before that row's variable reaches 0.

row	cste	x_3 column	ratio
x_4	2	3/2	2 ÷ (3/2) = **4/3**
x_5	3	5/2	3 ÷ (5/2) = **6/5**
x_6	2	3	2 ÷ 3 = **2/3** ← smallest
x_1	1	-1/2	skip: negative number (-1/2), so x_1 grows

The smallest ratio is **2/3**, in the x_6 row: **x_6 leaves** (it drops to 0). Pivot = the number where the x_3 column meets the x_6 row = **3**.

Table before the rewrite: x_3 column shaded, x_6 row outlined, pivot boxed

basis	x_1	x_2	x_3	x_4	x_5	x_6	x_7	cste
x_4	0	3/2	3/2	1	0	0	-1/2	2
x_5	0	3/2	5/2	0	1	0	1/2	3
x_6	0	-4	3	0	0	1	-1	2
x_1	1	3/2	-1/2	0	0	0	1/2	1
Z	0	-5/2	11/2	0	0	0	-5/2	-5

Step 3 Rewrite the table

(a) Divide the x_6 row by the pivot 3; it becomes the x_3 row. (b) In every other row, subtract (its number in the x_3 column) × the new pivot row, so the x_3 column becomes 1 in its own row and 0 everywhere else.

Table after round 2

basis	x_1	x_2	x_3	x_4	x_5	x_6	x_7	cste	how this row was computed
x_4	0	7/2	0	1	0	-1/2	0	1	old x_4 row − (3/2)·new pivot row
x_5	0	29/6	0	0	1	-5/6	4/3	4/3	old x_5 row − (5/2)·new pivot row
x_3	0	-4/3	1	0	0	1/3	-1/3	2/3	old x_6 row ÷ 3
x_1	1	5/6	0	0	0	1/6	1/3	4/3	old x_1 row + (1/2)·new pivot row
Z	0	29/6	0	0	0	-11/6	-2/3	-26/3	**old Z row − (11/2)·new pivot row**

Read the corner. In the basis column: x_4 = 1, x_5 = 4/3, x_3 = 2/3, x_1 = 4/3. Not in the basis, so equal to 0: x_2, x_6, x_7.

Corner (x_1, x_2, x_3) = **(4/3, 0, 2/3)**, profit Z = **26/3** (the Z-row cste −26/3 without its minus sign).

5 Round 3: x_2 enters, x_5 leaves

ROUND 3

Step 1 What to increase?

Z row (for variables not in the basis): 29/6 under x_2, −11/6 under x_6, −2/3 under x_7. The biggest positive number is **29/6**, under x_2: one more unit of x_2 adds 29/6 to the profit, the most. So **x_2 enters** (it will stop being 0).

Step 2 How far can x_2 go?

For each row with a positive number in the x_2 column: cste ÷ that number = how much x_2 fits before that row's variable reaches 0.

row	cste	x_2 column	ratio
x_4	1	7/2	1 ÷ (7/2) = **2/7**
x_5	4/3	29/6	4/3 ÷ (29/6) = **8/29** ← smallest
x_3	2/3	-4/3	skip: negative number (-4/3), so x_3 grows
x_1	4/3	5/6	4/3 ÷ (5/6) = **8/5**

The smallest ratio is **8/29**, in the x_5 row: **x_5 leaves** (it drops to 0). Pivot = the number where the x_2 column meets the x_5 row = **29/6**.

Table before the rewrite: x_2 column shaded, x_5 row outlined, pivot boxed

basis	x_1	x_2	x_3	x_4	x_5	x_6	x_7	cste
x_4	0	7/2	0	1	0	-1/2	0	1
x_5	0	29/6	0	0	1	-5/6	4/3	4/3
x_3	0	-4/3	1	0	0	1/3	-1/3	2/3
x_1	1	5/6	0	0	0	1/6	1/3	4/3
Z	0	29/6	0	0	0	-11/6	-2/3	-26/3

Step 3 Rewrite the table

(a) Divide the x_5 row by the pivot 29/6; it becomes the x_2 row. (b) In every other row, subtract (its number in the x_2 column) × the new pivot row, so the x_2 column becomes 1 in its own row and 0 everywhere else.

Table after round 3

basis	x_1	x_2	x_3	x_4	x_5	x_6	x_7	cste	how this row was computed
x_4	0	0	0	1	-21/29	3/29	-28/29	1/29	old x_4 row − (7/2)·new pivot row
x_2	0	1	0	0	6/29	-5/29	8/29	8/29	old x_5 row ÷ (29/6)
x_3	0	0	1	0	8/29	3/29	1/29	30/29	old x_3 row + (4/3)·new pivot row
x_1	1	0	0	0	-5/29	9/29	3/29	32/29	old x_1 row − (5/6)·new pivot row
Z	0	0	0	0	-1	-1	-2	-10	**old Z row − (29/6)·new pivot row**

Read the corner. In the basis column: x_4 = 1/29, x_2 = 8/29, x_3 = 30/29, x_1 = 32/29. Not in the basis, so equal to 0: x_5, x_6, x_7.

Corner (x_1, x_2, x_3) = **(32/29, 8/29, 30/29)**, profit Z = **10** (the Z-row cste −10 without its minus sign).

6 Stop? Read the answer

Z row (variables not in the basis): −1 under x_5, −1 under x_6, −2 under x_7. **No positive number**: any change would lower the profit, so we stop. This is the best plan.

No 0 under a variable outside the basis, so this best plan is the only one.

7 Check

Plug (32/29, 8/29, 30/29) in: 32/29 + 24/29 + 30/29 = 86/29 ≈ 2.97 ≤ 3 ✓ (x_4 = 1/29 left); −32/29 + 90/29 = 58/29 = 2 ✓ (full); 64/29 − 8/29 + 60/29 = 116/29 = 4 ✓ (full); 64/29 + 24/29 − 30/29 = 58/29 = 2 ✓ (full). Profit 5·32/29 + 5·8/29 + 3·30/29 = (160 + 40 + 90)/29 = 290/29 = **10**. Trying every corner gives the same maximum.

ANSWER

x_1 = 32/29, x_2 = 8/29, x_3 = 30/29, Z max = 10, after 3 rounds. Limits 2, 3 and 4 are full; the first has 1/29 left.

Exercise 6 Three rounds, a tie with a leftover variable

ASKED

(P): max $Z = 3x_1 + 2x_2 + 4x_3$ subject to $x_1 + x_2 + 2x_3 \le 4$, $2x_1 + 3x_2 \le 7$, $2x_1 + x_2 + 3x_3 \le 7$, all ≥ 0. 1) Standard form. 2) Solve with the simplex.

METHOD

Slacks, then the three steps. In round 3 the Z row shows 1 under x_2 and under the leftover x_4: break the tie by the bigger gain.

1 Question 1: standard form

One leftover per limit: x_4, x_5, x_6.

$$x_1 + x_2 + 2x_3 + x_4 = 4$$
$$2x_1 + 3x_2 + x_5 = 7$$
$$2x_1 + x_2 + 3x_3 + x_6 = 7$$
$$Z = 3x_1 + 2x_2 + 4x_3 \rightarrow \text{maximise} \cdot \text{all variables} \ge 0$$

2 Starting table

Starting table

basis	x_1	x_2	x_3	x_4	x_5	x_6	cste
x_4	1	1	2	1	0	0	4
x_5	2	3	0	0	1	0	7
x_6	2	1	3	0	0	1	7
Z	3	2	4	0	0	0	0

Read the corner. In the basis column: $x_4 = 4$, $x_5 = 7$, $x_6 = 7$. Not in the basis, so equal to 0: x_1, x_2, x_3.

Corner (x_1, x_2, x_3) = **(0, 0, 0)**, profit Z = **0** (the Z-row cste 0 without its minus sign).

3 Round 1: x_3 enters, x_4 leaves

ROUND 1

Step 1 **What to increase?**

Z row (for variables not in the basis): 3 under x_1, 2 under x_2, 4 under x_3. The biggest positive number is **4**, under x_3: one more unit of x_3 adds 4 to the profit, the most. So **x_3 enters** (it will stop being 0).

Step 2 **How far can x_3 go?**

For each row with a positive number in the x_3 column: cste ÷ that number = how much x_3 fits before that row's variable reaches 0.

row	cste	x_3 column	ratio
x_4	4	2	4 ÷ 2 = **2** ← smallest
x_5	7	0	skip: 0 in the column (never runs out)
x_6	7	3	7 ÷ 3 = **7/3**

The smallest ratio is **2**, in the x_4 row: **x_4 leaves** (it drops to 0). Pivot = the number where the x_3 column meets the x_4 row = **2**.

Table before the rewrite: x_3 column shaded, x_4 row outlined, pivot boxed

basis	x_1	x_2	x_3	x_4	x_5	x_6	cste
x_4	1	1	2	1	0	0	4
x_5	2	3	0	0	1	0	7
x_6	2	1	3	0	0	1	7
Z	3	2	4	0	0	0	0

Step 3 **Rewrite the table**

(a) Divide the x_4 row by the pivot 2; it becomes the x_3 row. (b) In every other row, subtract (its number in the x_3 column) × the new pivot row, so the x_3 column becomes 1 in its own row and 0 everywhere else.

Table after round 1

basis	x_1	x_2	x_3	x_4	x_5	x_6	cste	how this row was computed
x_3	1/2	1/2	1	1/2	0	0	2	old x_4 row ÷ 2
x_5	2	3	0	0	1	0	7	copied (0 in the column)
x_6	1/2	-1/2	0	-3/2	0	1	1	old x_6 row − 3·new pivot row
Z	1	0	0	-2	0	0	-8	old Z row − 4·new pivot row

Read the corner. In the basis column: x_3 = 2, x_5 = 7, x_6 = 1. Not in the basis, so equal to 0: x_1, x_2, x_4.

Corner (x_1, x_2, x_3) = **(0, 0, 2)**, profit Z = **8** (the Z-row cste −8 without its minus sign).

4 Round 2: x_1 enters, x_6 leaves

ROUND 2

Step 1 **What to increase?**

Z row (for variables not in the basis): 1 under x_1, 0 under x_2, −2 under x_4. The biggest positive number is **1**, under x_1: one more unit of x_1 adds 1 to the profit, the most. So **x_1 enters** (it will stop being 0).

Step 2 **How far can x_1 go?**

For each row with a positive number in the x_1 column: cste ÷ that number = how much x_1 fits before that row's variable reaches 0.

row	cste	x_1 column	ratio
x_3	2	1/2	2 ÷ (1/2) = **4**
x_5	7	2	7 ÷ 2 = **7/2**
x_6	1	1/2	1 ÷ (1/2) = **2** ← smallest

The smallest ratio is **2**, in the x_6 row: **x_6 leaves** (it drops to 0). Pivot = the number where the x_1 column meets the x_6 row = **1/2**.

Table before the rewrite: x_1 column shaded, x_6 row outlined, pivot boxed

basis	x_1	x_2	x_3	x_4	x_5	x_6	cste
x_3	1/2	1/2	1	1/2	0	0	2
x_5	2	3	0	0	1	0	7
x_6	1/2	-1/2	0	-3/2	0	1	1
Z	1	0	0	-2	0	0	-8

Step 3 **Rewrite the table**

(a) Divide the x_6 row by the pivot 1/2; it becomes the x_1 row. (b) In every other row, subtract (its number in the x_1 column) × the new pivot row, so the x_1 column becomes 1 in its own row and 0 everywhere else.

Table after round 2

basis	x_1	x_2	x_3	x_4	x_5	x_6	cste	how this row was computed
x_3	0	1	1	2	0	-1	1	old x_3 row − (1/2)·new pivot row
x_5	0	5	0	6	1	-4	3	old x_5 row − 2·new pivot row
x_1	1	-1	0	-3	0	2	2	old x_6 row ÷ (1/2)
Z	0	1	0	1	0	-2	-10	**old Z row − new pivot row**

Read the corner. In the basis column: x_3 = 1, x_5 = 3, x_1 = 2. Not in the basis, so equal to 0: x_2, x_4, x_6.

Corner (x_1, x_2, x_3) = **(2, 0, 1)**, profit Z = **10** (the Z-row cste −10 without its minus sign).

5 Round 3: x_2 enters, x_5 leaves

ROUND 3

Step 1 **What to increase?**

Z row (for variables not in the basis): 1 under x_2, 1 under x_4, −2 under x_6. **Tie:** the biggest positive number, 1, appears more than once. Do step 2 for each tied variable and compare the profit each would bring: x_2: smallest ratio 3/5, so profit gained 1 × 3/5 = **3/5**; x_4: smallest ratio 1/2, so profit gained 1 × 1/2 = **1/2**. The bigger gain wins: **x_2 enters**.

Step 2 **How far can x_2 go?**

For each row with a positive number in the x_2 column: cste ÷ that number = how much x_2 fits before that row's variable reaches 0.

row	cste	x_2 column	ratio
x_3	1	1	1 ÷ 1 = **1**
x_5	3	5	3 ÷ 5 = **3/5** ← smallest
x_1	2	-1	skip: negative number (-1), so x_1 grows

The smallest ratio is **3/5**, in the x_5 row: **x_5 leaves** (it drops to 0). Pivot = the number where the x_2 column meets the x_5 row = **5**.

Table before the rewrite: x_2 column shaded, x_5 row outlined, pivot boxed

basis	x_1	x_2	x_3	x_4	x_5	x_6	cste
x_3	0	1	1	2	0	-1	1
x_5	0	5	0	6	1	-4	3
x_1	1	-1	0	-3	0	2	2
Z	0	1	0	1	0	-2	-10

Step 3 **Rewrite the table**

(a) Divide the x_5 row by the pivot 5; it becomes the x_2 row. (b) In every other row, subtract (its number in the x_2 column) × the new pivot row, so the x_2 column becomes 1 in its own row and 0 everywhere else.

Table after round 3

basis	x_1	x_2	x_3	x_4	x_5	x_6	cste	how this row was computed
x_3	0	0	1	4/5	-1/5	-1/5	2/5	old x_3 row − new pivot row
x_2	0	1	0	6/5	1/5	-4/5	3/5	old x_5 row ÷ 5
x_1	1	0	0	-9/5	1/5	6/5	13/5	old x_1 row + new pivot row
Z	0	0	0	-1/5	-1/5	-6/5	-53/5	**old Z row − new pivot row**

Read the corner. In the basis column: x_3 = 2/5, x_2 = 3/5, x_1 = 13/5. Not in the basis, so equal to 0: x_4, x_5, x_6.

Corner (x_1, x_2, x_3) = **(13/5, 3/5, 2/5)**, profit Z = **53/5** (the Z-row cste −53/5 without its minus sign).

6 Stop? Read the answer

Z row (variables not in the basis): −1/5 under x_4, −1/5 under x_5, −6/5 under x_6. **No positive number**: any change would lower the profit, so we stop. This is the best plan.

No 0 under a variable outside the basis, so this best plan is the only one.

7 Check

Plug (13/5, 3/5, 2/5) in: 13/5 + 3/5 + 4/5 = 20/5 = 4 ✓ (full); 26/5 + 9/5 = 35/5 = 7 ✓ (full); 26/5 + 3/5 + 6/5 = 35/5 = 7 ✓ (full). Profit 3·13/5 + 2·3/5 + 4·2/5 = (39 + 6 + 8)/5 = **53/5 = 10.6**. Trying every corner gives the same maximum.

ANSWER

1) $x_1 + x_2 + 2x_3 + x_4 = 4$, $2x_1 + 3x_2 + x_5 = 7$, $2x_1 + x_2 + 3x_3 + x_6 = 7$, all ≥ 0.

2) **x_1 = 13/5, x_2 = 3/5, x_3 = 2/5, Z max = 53/5 = 10.6**, after 3 rounds. All three limits are full.